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Github: <https://github.com/Meet-6226/Flutter-Assignment>

What I learn

While working on this Flutter assignment, I learned how to fetch data from a **public REST API** and display it inside a Flutter application. I used a public API such as JSONPlaceholder to understand how real applications communicate with online servers and receive data. This helped me understand the basic concept of APIs and how information such as posts or user details can be loaded dynamically instead of being written directly inside the application.

I learned how to make an API request using Flutter and handle the response received from the server. I also understood the importance of converting the JSON response into a format that can be easily used in the application. This helped me get a basic understanding of how JSON data is structured and how Flutter can work with data received from an external source.

Another important concept I learned was the **FutureBuilder** widget. I used FutureBuilder to handle the different states of an API request, such as loading, successful data retrieval, and errors. This helped me understand asynchronous programming in Flutter because the API does not return the data immediately. Instead, the application has to wait for the response while still keeping the user interface responsive.

I also learned how to use **SharedPreferences** to store data locally on the device. In this assignment, I used it to cache the last successfully fetched API result. This means that if the API is unavailable or there is no internet connection, the application can use the previously saved data instead of showing an empty screen. This helped me understand the difference between fetching fresh data from an API and storing data locally for later use.

Overall, this assignment helped me understand some important concepts used in real-world Flutter applications, including **REST APIs, JSON data, asynchronous operations, FutureBuilder, and local data storage using SharedPreferences**. I also learned that handling loading and error states is important when working with APIs. This assignment gave me a better understanding of how Flutter applications can fetch, display, and temporarily store real data.

